Distinguishing magma recharge from crustal memory during repeated volcanic unrest

Joint inference, repeated withdrawals, and the design of geodetic observations at Svartsengi

> **Status** This is a prospective scientific draft. Hypotheses, acceptance gates, and planned figures are not completed results. The detailed experiment specification is maintained in `DIKE_MEMORY_PAPER_PLAN.md`.

1 Abstract

Surface deformation during volcanic unrest reflects magma storage, magma compressibility, and the time-dependent mechanical response of the surrounding crust. A declining uplift rate can therefore admit several interpretations, including pressure-dependent recharge and relaxation following earlier withdrawals. We propose to determine when geodetic observations can distinguish these processes at a repeatedly recharging magma reservoir. Plane-strain benchmarks and synthetic inference will establish the workflow before selected tests are repeated in 3D. A verified chamber model will couple mass balance to viscoelastic deformation, with elastic, constant-recharge, and pressure-feedback alternatives tested on common synthetic observations. Supply will be inferred jointly with uncertain rheology, compressibility, withdrawal sizes, and initial mechanical memory. Repeated withdrawals will be evaluated as probes of the system response, with chronological prediction of withheld cycles providing a test beyond fitting observed histories. Observing strategies will be judged by actual supply error, uncertainty coverage, and detection delay under model mismatch. Svartsengi provides the motivating case, but a site-specific attribution will require the processed displacement observations. The intended outcome is a map of recoverable and ambiguous regimes, including cases where additional stations cannot resolve the supply-memory tradeoff.

2 Scientific motivation and contribution

Viscoelastic memory is established physics. Segall showed that some compressibility regimes permit post-eruption reinflation without recharge.[1] Liao and colleagues explicitly demonstrated history dependence and frequency-dependent response around thermally structured magma reservoirs.[2] Recent 3D models also quantify departures from elastic interpretations when feeding, thermal weakening, and magma compressibility interact.[3]

The proposed advance is to test recovery of supply when those physical properties are uncertain. The central hypothesis is that repeated withdrawals of different sizes and spacings constrain shared response properties sufficiently to improve subsequent supply estimates. A useful alternative outcome is a defensible bound on what geodesy alone cannot determine. Neither uniqueness nor an improved station layout is assumed in advance.

3 Physical and observational model

3.1 Separate mass, chamber expansion, and apparent source volume

The initial model is a well-mixed compressible chamber embedded in a Maxwell shell and an elastic host. Temperature is prescribed and deformation is small. The shell represents effective mechanical weakening; it does not reproduce hydraulic transport through a porous mush.[6] For fixed temperature and composition and small density perturbations,

$$\dot{m}_{\mathrm{in}} - \dot{m}_{\mathrm{out}} = \rho \dot{V} + \rho \beta_m V \dot{p}. \tag{1}$$

Defining both volumetric fluxes at the same reference density gives

$$Q_{\mathrm{in}} - Q_{\mathrm{out}} = \dot{V} + \beta_m V \dot{p}. \tag{2}$$

The model must distinguish geometric chamber volume, injected reference volume, pressure, and volume inferred by an elastic source inversion. A constant pressure load and a constant injection rate are different experiments. Increased compliance can produce more displacement per unit pressure while reducing pressure generated per unit injection; these effects can partly cancel.[3]

We will compare constant recharge with pressure-dependent feeding,

$$Q_{\mathrm{in}} = C_f \left(p_{\mathrm{deep}} - p \right), \tag{3}$$

and with externally imposed persistent supply changes. A decline in inflow under pressure feedback does not alone imply a decline in deep-source pressure or conductivity. Source controls and observed withdrawal estimates will enter with consistent units and uncertainties.

3.2 Observations and inference parameters

Daily GNSS displacement is the primary observable. The initial target is mean positive recharge over the preceding 30 days, with a persistent 20% change used as a pilot detection experiment. These choices will be frozen before production and checked against other windows. Horizontal and vertical components will be compared with appropriate covariance. Any velocity experiment must propagate the covariance introduced by differentiation.

Positive recharge is represented in coarse time bins or through a correlated prior; signed withdrawals are separate parameters. Shared nuisance parameters include shell viscosity and thickness, magma compressibility, source geometry, and low-dimensional modes of initial stress history. Their uncertainty must be propagated into supply estimates. Correlation lengths and parameter scales will be declared so that prior smoothing is not mistaken for information from data.

3.3 Temporal kernels and the limits of spatial sampling

For positive recharge bins, let $q_j = \ln\left(\frac{Q_j}{Q_{\mathrm{ref}}}\right)$. The discrete Jacobian $K_{ij} = \frac{\partial u_i}{\partial q_j}$ has displacement units, with each bin's duration included in the forward calculation. A continuous functional derivative is a density in time and has different units. A finite step perturbation measures an integrated response, approaching a sum of linear kernels for small perturbations.

If $u(x,t) \approx g(x)a(t)$, additional stations can measure the same temporal mixture more precisely without adding an independent spatial signature of memory. We will inspect noise-whitened, parameter-scaled singular values and nuisance correlations, then test actual inversions. Local rank is a diagnostic, not proof of global uniqueness. Marginal supply uncertainty must integrate out rheological uncertainty rather than condition on its true value.

4 Experiment sequence and decision gates

4.1 S0: qualify the numerical memory and derivative

The existing dike step experiments are exploratory. Their delay must be tested with plasticity disabled, with fixed physical viscoplastic viscosity, and with regularization tied to the time step. The last sequence changes the constitutive model as well as the discretization. Persistence at fixed viscosity establishes neither a natural-crust parameter value nor a rate-independent plastic limit.

Time, mesh, source footprint, domain, and solver tolerance studies will evaluate the difference of paired runs at common physical times. Perturbations of both signs and several magnitudes will test local linearity. Controlled initial states will separate spin-up from the response being measured.

Compressible-history gradients require an independent full-system transpose check, including storage, augmentation, pressure objectives, and accepted history. The present working tree contains relevant solver changes; earlier finite-difference agreement is not certification of the current implementation. Frozen thermal, pore-pressure, lithostatic, or geometric fields must also be frozen in the finite-difference oracle. Fully coupled derivatives require their dependence.

> **Proposed numerical gate.** Successive refinements change the response by less than 2%, and numerical uncertainty is below 10% of the smallest claimed physical effect. Nonzero gradients agree within 1% of a finite-difference plateau with consistent Taylor behavior. These are acceptance targets, not achieved results.

4.2 S1: benchmark the plane-strain chamber before inference

The first model is deliberately two-dimensional plane strain. Existing circular and elliptical cavity solutions and the Mogi line-source limit will test elastic response, with boundary and finite-source errors separated. Discrete source, chamber expansion, pressure storage, and accepted stress history will be checked on a fixed mesh before adding thermal evolution or yielding. Fluxes are areas per time and cumulative injections are areas per unit strike.

Plane strain is the primary setting for cheap benchmarks and synthetic inference, not a quantitatively realistic Svartsengi model. It changes spatial pattern, compliance, and time response relative to 3D. Those differences do not block synthetic tests of mechanics and identifiability. Once that workflow is settled, a spherical benchmark against the applicable Segall solution and selected 3D experiments will assess transfer to more realistic geometry.[1]

4.3 S2: determine what can be recovered

Three families form the initial comparison: elastic crust with pressure-dependent recharge (E-P), viscoelastic crust with constant recharge (V-C), and viscoelastic crust with pressure-dependent recharge (V-P). No-recharge and known-rheology controls isolate mechanisms. A first-order corrected-elastic baseline will be included where its assumptions apply.[3]

One cycle will be compared with repeated withdrawals having different magnitudes and repose intervals. Joint estimation will include withdrawal uncertainty and initial memory. Regimes will be classified as data-resolved, prior-dominated, or unresolved. Flexible supply histories can absorb a wrong mechanical model; therefore good fits alone will not establish the correct mechanism.

5 Recovery, prediction, and observing-system design

5.1 S3: distinguish fitting from prediction

Independent synthetic realizations will cover constant, pressure-controlled, and externally changed supply. Competing models will be fitted with uncertain rheology and deliberately wrong geometry or viscosity. Supply bias, root mean square error, interval coverage, change-detection delay, false alarms, and predictive scores will be reported. The pilot starts with deterministic 2D finite-element tests. Once qualified, cached linear responses or a verified reduced model will support 100 noise realizations, checked against independent FEM runs. Sampling uncertainty will determine whether more realizations are needed.

Shared rheology will be estimated from training cycles. Leave-one-cycle-out reconstruction will be distinguished from chronological prediction using only past observations. Forecasts cannot use the true future withdrawal or future displacements. Tests supplied with those quantities are conditional reconstructions. Repeated events will be considered informative only if actual joint recovery or prediction improves, with uncertainty coverage remaining credible.

5.2 S4: optimize the decision-relevant observations

Observing strategies will compare existing or even spacing, maximum signal, elastic-model designs, and designs that marginalize uncertain physics. The primary objective is recent-supply uncertainty or error. D-optimal information is a secondary diagnostic. In a Gaussian linear model, joint expected information gain in nats is

$$I = \frac{1}{2} \ln \det \left(\boldsymbol{I} + C_{\text{prior}}^{\frac{1}{2}} J^T C_{\text{noise}}^{-1} J C_{\text{prior}}^{\frac{1}{2}} \right). \tag{4}$$

An unhalved log determinant is a different score. Supply-specific information must integrate out nuisance parameters consistently. Neither score measures the bias of an estimator using a wrong forward model. Selected designs will therefore be evaluated by actual inversions of independent noisy data.

Added stations will be compared with observation timing around withdrawals and with displacement-component choice at a declared observing cost. For continuous GNSS, better use of existing daily epochs is distinct from acquiring more data. Robustness tests will include uncertain geometry/rheology, correlated errors, and station outages. Feasible locations matter once the real network is used. Spatial configuration already has established sensitivity to topography and heterogeneity; Bayesian volcano design also has precedents in seismic monitoring. These motivate, rather than establish, a recharge-design advantage.[8, 9]

5.3 S5: summarize regimes without overclaiming forecasting

Selected slices will vary cycle length relative to nominal Maxwell time, hydraulic recharge relative to relaxation time, relative magma/chamber compressibility, shell size, source depth, and signal-to-noise. Instantaneous elastic chamber compliance defines the reference chamber compressibility. Effective coupled decay times will also be measured. A distributed thermal-viscosity profile will test the limits of a single-shell description.[2]

Supply recovery, volume recovery, pressure recovery, and eruption timing are different quantities. The initial forecast diagnostic is crossing a declared proxy threshold with threshold uncertainty. Prescribing event times cannot validate an eruption forecast. A failure and propagation model would need its own verification before supporting that claim.

6 Svartsengi application and planned evidence

Parks and colleagues infer repeated magma-domain inflation and deflation using GNSS and InSAR. Their automated volume history uses fixed source geometry because inflation is approximately self-similar.[4] The published volume series is a model-derived observable, not independent ground truth for magma flux. A 2026 conference abstract compares multiple elastic geometries and reports more stable source estimates after March 2024.[5] Early and later cycles will therefore be analyzed separately before assuming shared geometry throughout. The abstract is preliminary evidence.

A site-specific attribution needs processed three-component GNSS series, covariances, reference-frame and station histories, and event/source estimates with uncertainty. InSAR can constrain spatial patterns. The existing digitized volume and rate curves are useful for reproducing the published elastic summary, but cannot determine the physical cause of its curvature. Normalized spatial-pattern evolution and full-field residuals will be tested alongside time dependence. Tectonic motion, fault/dike deformation, and geothermal effects will be treated where relevant. Elastic withdrawal estimates cannot be imposed as exact physical magma losses in a test of elastic-model bias.

Figure	Scientific question	Required evidence
1	What does surface motion measure?	Verified budget; apparent versus physical volume; chamber response.
2	Can supply and memory be separated?	Joint sensitivities and competing fits, including an ambiguous case.
3	Does repeated-event inference generalize?	Actual recovery, interval coverage, withheld-cycle prediction.
4	Which observations improve recent supply?	Equal-cost design comparisons under uncertain and wrong physics.
5	Where is recovery reliable?	Dimensionless slices, noise dependence, and limited forecast proxies.

6.1 Interpretation, scope, and reproducibility

Thermal state is uncertain, and current unrest duration does not determine the thermal maturity of the plumbing system.[7] A mechanical shell also omits hydraulic mush processes. Benchmarks and synthetic tests will be completed in plane strain first; 3D follows once the mechanics and inference workflow are settled. A site-specific interpretation requires the later geometry assessment. The first paper will use the simplest verified model that addresses the supply-memory ambiguity. Yielding, evolving thermal fields, background strain, and broad dike-dip sweeps are targeted extensions, not prerequisites. Existing dike results remain historical exploratory evidence until their numerical and constitutive gates pass.

No result presently establishes that relaxation explains the Svartsengi supply trend or that a new station layout improves its inference. If the observations cannot distinguish mechanisms, the paper should report alternative histories, prior dependence, and useful supply bounds. A synthetic identifiability study remains possible without a real-data attribution. Journal choice and a methods companion should follow the demonstrated results.

The archive will include code revision and dirty diff, environments, meshes, source normalization, accepted time grids, initial histories, convergence and budget diagnostics, random seeds, prior/noise definitions, inversion settings, and figure scripts. The next executable step is the fixed-mesh linear plane-strain chamber pilot; a production network-design campaign follows only after identifiability.

Checked references

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> **Reading scope** This focused primary-source review positions the proposal; it does not establish exhaustive coverage or priority. No chamber inversion, real-data attribution, or observing-system improvement is claimed as completed.